

Name: \_\_\_\_\_

Student ID: \_\_\_\_\_

Signature: \_\_\_\_\_

UNSW, Sydney  
School of Electrical Engineering & Telecommunications

## FINAL EXAMINATION

Term 1, 2023

### ELEC3115 - Electromagnetic Engineering

1. Time allowed: 2 (two) hours
2. Reading Time: 10 (ten) minutes.
3. Total number of questions set in this paper: 5 (five). Candidates must answer **ALL five** questions: **THREE** from Part A and **TWO** from Part B.
4. Marks for each question are indicated at the right-hand side of each question. This examination paper carries 100 marks; it contributes 50% of the total course assessment.
5. This paper contains 10 pages.
6. Answer each question in a **separate answer book**. Write your name, student ID and Question number on the front page of **each answer book**.
7. This paper **MAY NOT** be retained by the candidate.
8. This is a closed book examination. Students may only bring the following materials into the examination room:
  - UNSW-approved electronic calculators.
  - Drawing instruments.
9. The Examination Section will supply two Smith charts.
10. Assumptions made in answering the questions should be stated explicitly.
11. All answers must be written in ink. Except where they are expressly required, pencils may only be used for drawing, sketching or graphical work.

## PART A

### QUESTION 1

**[20 marks]**

A two-wire overhead power transmission line has two identical but oppositely charged copper conductors as indicated in Figure 1. Each conductor has a circular cross-section of radius  $a = 5$  mm and the centre-to-centre separation between the conductors is  $d = 5$  cm. The conductors are bare and surrounded by air. [Given,  $\epsilon_0 = 8.854 \times 10^{-12}$  F/m]

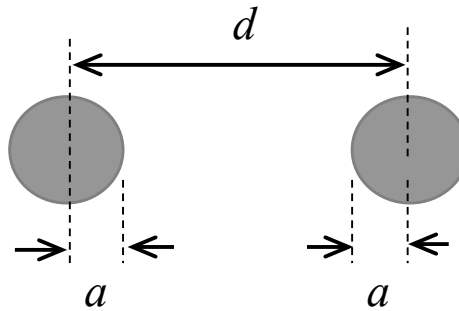


Figure 1

- (a) **[5 marks]** Show that when  $d \gg a$ , the amplitude of the electric field at an arbitrary radial distance  $r$  along the shortest line joining the centres of the two conductor is given by

$$|E(r)| = \frac{V}{2 \ln \frac{d}{a}} \left( \frac{1}{r} + \frac{1}{d-r} \right) \text{ V/m,}$$

where potential difference between the conductors is  $V = \left( \frac{\rho_l}{\pi \epsilon_0} \ln \frac{d}{a} \right)$  Volt and  $\rho_l$  is the line charge density in C/m.

- (b) **[4 marks]** Sketch the variation of the amplitude of the electric field along the shortest line joining the conductors indicating where the electric field is maximum and minimum for a potential difference between the two conductors  $V = 50$  kV. You must show the numerical values of maximum and minimum electric fields on your sketch.

- (c) **[5 marks]** Assuming  $d \gg a$ , show that capacitance between the two conductor per unit length is  $C \approx \frac{\pi \epsilon_0}{\ln \frac{d}{a}}$  F/m.

- (d) **[4 marks]** Since the capacitance found using (c) is an approximation, let us refer to it as  $C_{\text{approx}}$ . The same capacitance when found using *method of image* is more accurate and given as,

$$C_{\text{actual}} = \frac{\pi \epsilon_0}{\ln \left[ \left( \frac{d}{2a} \right) + \sqrt{\left( \frac{d}{2a} \right)^2 - 1} \right]} \text{ F/m.}$$

Find the percentage (%) error between  $C_{\text{approx}}$  and  $C_{\text{actual}}$  when  $d = 5$  cm and 1.1 cm.

- (e) **[2 marks]** Comparing the percentage (%) error found in (d), we can conclude that if centre-to-centre separation  $d$  reduces, error increases – explain why.

**QUESTION 2****[20 marks]**

- (a) [8 marks] A co-axial cable of Figure 2 has a conductor of radius  $a$  and the radial distance from the centre of the conductor to the metallic shield is  $b$ . The potential difference between the conductor and the metallic shield is  $V_0$ . The cable is assumed to be very long. Find the electric field distribution  $E$  between the conductor and metallic shield using Laplace's equation

$$\bar{\nabla}^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \theta^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

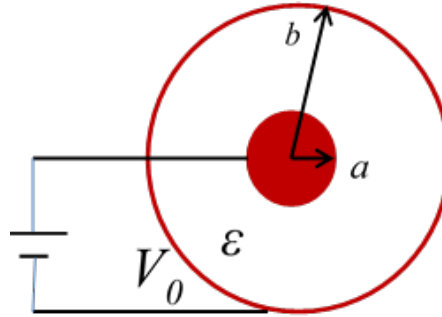


Figure 2

- (b) [2 marks] The coaxial cable of Figure 2 is used to supply power to an amplifier. The inner conductor of the cable has the radius  $a = 0.5$  mm and the outer radius  $b = 5$  mm.  $V_0 = 60$  V DC is connected between the two conductors with positive end at the inner conductor, find the maximum and minimum values of the electric field  $E$  in the dielectric material between the two conductors.
- (c) [6 marks] Consider the material between the inner and the outer conductors of the co-axial cable of Figure 2 is an imperfect dielectric with a conductivity  $\sigma$ . Show that the leakage resistance in the dielectric material between the inner and outer cylindrical surfaces is given by

$$R_{lk} = \frac{1}{2\pi l \sigma} \ln \frac{b}{a}$$

where  $l$  is the length of the cable along the  $z$ -axis of the cylindrical co-ordinate.

- (d) [4 marks] If the length of the cable 100 m and the conductivity of the imperfect dielectric  $\sigma = 5 \times 10^{-8}$  S/m. Find the leakage current and the power loss between the inner and outer surface when supplying the power to the amplifier with  $V_0 = 60$  V.

### QUESTION 3

[20 marks]

(a) [4 marks] Describe how *virtual displacement* can be used to calculate force between magnetic circuits, assuming that the displacement takes place at constant current.

(b) [9 marks] An electromagnetically activated piston -pump is shown in Figure 3(a). The Figure 3(b) shows the detailed dimensions of the electromagnetic core of the pump. The coil contains  $N=5000$  turns and carries a current  $I = 0.1\text{A}$ . Permeability of the core iron is  $\mu = 1000\mu_0$  H/m and the air gap is  $l_g = 5\text{ mm}$  long. Assume no leakage and fringing at the air gap. Using *virtual displacement* method, calculate the force  $F$  exerted by the moving piece on the piston. [Given,  $\mu_0 = 4\pi \times 10^{-7}$  H/m].

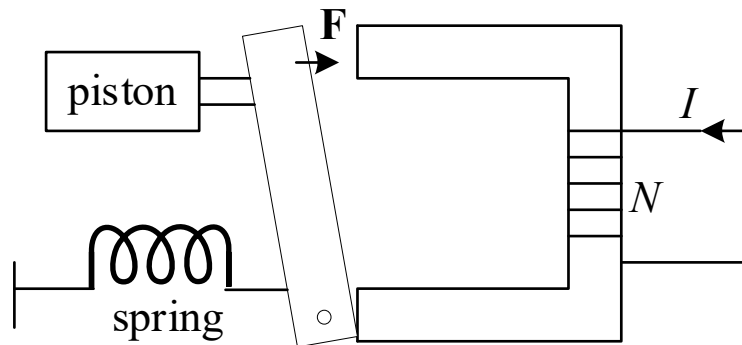


Figure 3 (a)

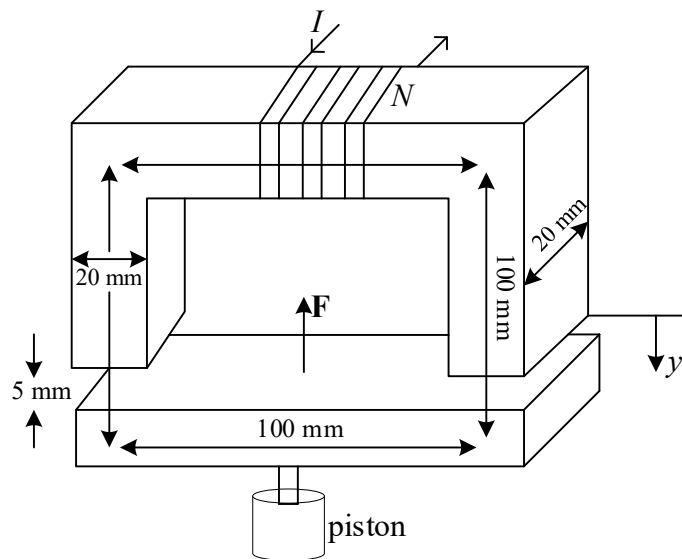


Figure 3 (b)

- (c) [2 marks] Assuming uniform flux density in the core, calculate the self-inductance of the coil of the electromagnet of the pump.
- (d) [2 marks] If the coil is supplied with 50 Hz AC voltage and current  $I = 0.5\text{ A}$  in (b) is the peak value, neglecting coil resistance, calculate the rms value of the supply voltage.
- (e) [3 marks] AC supply causes eddy current loss in the core of the pump. If the eddy current loss co-efficient is  $K_e = 0.12$ , calculate the total eddy current loss in Watts for the 50Hz AC supply. What steps could be taken to reduce eddy current loss in magnetic devices?

See next page for Part B questions

## Part B

### QUESTION 4

[24 marks]

Consider a lossless coaxial cable of length  $l = 1$  m, filled by a non-magnetic dielectric.

The cable possesses an inductance per unit length  $\mathcal{L} = 250$  nH/m and a capacitance per unit length  $\mathcal{C} = 100$  pF/m.

The cable is terminated by a short circuit, and fed by an ideal DC voltage generator with zero internal resistance and voltage  $V_G = 1$  V. An ideal switch is interposed between the generator and the cable, as shown in Figure 4 below.

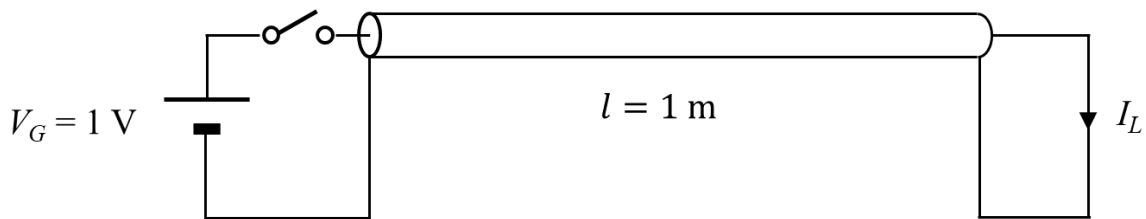


Figure 4

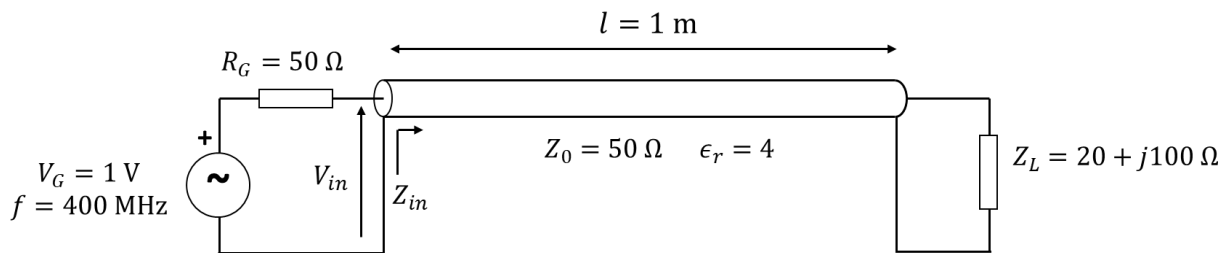
At time  $t = 0$ , the switch is closed and the generator is connected to the cable. For the time interval  $t = 0 - 100$  ns draw:

- (i) The reflection diagram, which must contain the numerical values of the voltage and current on every path, and a correct labelling of the time axis. [8 marks]
- (ii) The current circulating in the short-circuit termination,  $I_L(t)$ . [8 marks]
- (iii) Construct a lumped-element circuit that best approximates the behaviour of the “real” circuit. Describe the reasoning you use to construct this circuit and provide the numerical value of its parameters. [4 marks]
- (iv) Draw the current  $I_L(t)$  in the time interval  $t = 0 - 100$  ns as would be obtained from the lumped-elements approximation drawn at point (iii) above. Discuss the difference or similarity between the “true” current drawn at point (ii), and the one you obtain here. [4 marks]

See next page for question 5

**QUESTION 5****[16 marks]**

Consider a coaxial cable, filled with a non-magnetic dielectric with  $\epsilon_r = 4$ . With this dielectric, the cable has a characteristic impedance  $Z_0 = 50 \Omega$ . The cable is 1 m long, terminated on the right by an impedance  $Z_L = 20 + j100 \Omega$ , and connected on the left to a voltage generator with internal resistance  $R_G = 50 \Omega$  that produces an AC voltage of amplitude  $V_G = 1 \text{ V}$  and frequency  $f = 400 \text{ MHz}$ .



- (i) Calculate the wavelength  $\lambda$  of the propagating signal in the cable. **[3 marks]**
- (ii) Calculate the voltage standing wave ratio VSWR in the cable. **[3 marks]**
- (iii) Using the supplied Smith chart, calculate the input impedance  $Z_{in}$  seen by the generator at the entrance of the cable (to the right of its own internal resistance). Provide a clear written explanation of the steps you take while using the Smith chart to arrive at the result, and attach the completed Smith chart to your exam paper. **[8 marks]**
- (iv) Calculate the voltage  $V_{in}$  at the input of the cable. **[2 marks]**

**END OF EXAMINATION PAPER**

# Par A Formula Sheet

## Useful Constants:

Speed of light  $c = 3 \times 10^8 \text{ m/s}$

Permittivity of free space  $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$

Permeability of free space  $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$

Electrons  $e^- = -1.60 \times 10^{-19} \text{ C}$

## Electrostatic

### Postulate 1: Gauss Law

$$\oint_S \vec{E} \cdot d\vec{s} = \frac{Q_{\text{enclosed}}}{\epsilon}$$

$$\vec{D} = \epsilon \vec{E} = \frac{Q_{\text{enclosed}}}{s}$$

$$q = \int_v \rho_v dv \text{ or } \int_s \rho_s ds \text{ or } \int_l \rho_l dl$$

### Postulate 2: Electrical Potential

$$V_{ba} = - \int_a^b \vec{E} \cdot d\vec{l} = Ed$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} = \frac{1}{\epsilon_0} (\rho + \rho_p)$$

$$\rho_p = -\vec{\nabla} \cdot \vec{P}$$

### Case 1: Charge

$$\vec{E} = \frac{Q}{4\pi\epsilon R^2} \vec{a}_r$$

$$V = \frac{q}{4\pi\epsilon R}$$

### Case 2: Sphere (Charges are uniformly distributed)

$$\vec{E}_{\text{inside}} = \frac{\rho_v R}{3\epsilon} \vec{a}_r$$

$$\vec{E}_{\text{outside}} = \frac{\rho_v b^3}{3\epsilon R^2} \vec{a}_r$$

### Case 3: Cylindrical body (Any influence of ground is negligible)

$$\vec{E} = \frac{\rho_l}{2\pi\epsilon R} \vec{a}_r$$

$$V_{12} = \frac{\rho_l}{2\pi\epsilon} \ln \frac{r_1}{r_2}$$

## Capacitance

$$C = \frac{Q}{V} = \frac{\epsilon A}{d}$$

$$V = \frac{Qd}{\epsilon A}$$

### Case 1: Capacitance between 2 parallel conductors (transmission line) carrying current in opposite directions

$$C \approx \frac{\pi\epsilon}{\ln \frac{d-a}{a}} [\text{F/m}]$$

### Case 2: Capacitance between the conductor and the metallic shield

$$C = \frac{\rho_l}{V_{ab}} = \frac{2\pi\epsilon}{\ln \frac{b}{a}} [\text{F/m}]$$

### Case 3: Capacitance of a multi-conductor system (Capacitance matrix)

$$\begin{bmatrix} Q_1 \\ Q_2 \\ Q_3 \end{bmatrix} = \begin{bmatrix} C_{10} + C_{12} + C_{13} & -C_{12} & -C_{13} \\ -C_{12} & C_{20} + C_{12} + C_{23} & -C_{23} \\ -C_{13} & -C_{23} & C_{30} + C_{13} + C_{23} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

- $\vec{E}_n = \frac{\rho_s}{\epsilon}, \vec{D}_n = \rho_s$
- $\vec{E}_t = 0, \vec{D}_t = 0$

## Insulator

- Point vector  $\vec{p} = q\vec{d}$
- Polarization  $\vec{P} = \vec{D} - \epsilon_0 \vec{E}$
- $|\vec{D}_{1n}| = |\vec{D}_{2n}|$
- $|\vec{E}_{1t}| = |\vec{E}_{2t}|$

### Case: Electric field at the interface of 2 dielectric

$$E_{\text{max}} [m1] = \frac{\rho_l}{2\pi\epsilon_2} [\text{Find } \rho_l]$$

$$E_{\text{max}} [m2] = \frac{\rho_l}{2\pi\epsilon_1} [\text{Find } r]$$

$$\epsilon_1 E_{\text{max}} [m1] = \epsilon_2 E_{\text{max}} [m2]$$

$$\text{Maximum working voltage} = \frac{-\rho_l}{2\pi\epsilon} \ln \frac{d}{r}$$

$$\text{Total working voltage} = \frac{-\rho_l}{2\pi\epsilon_0} \left[ \frac{1}{\epsilon_{rp}} \ln \frac{r_o}{r_p} + \frac{1}{\epsilon_{rr}} \ln \frac{r_p}{r_r} \right]$$

## Current

- Current density  $\vec{J} = \frac{I}{S} = \rho \frac{\Delta l}{\Delta t} [\text{A/m}^2]$
- Conductivity of medium  $\sigma = \rho \mu_e [\text{A/V-m}] [\text{S/m}]$
- Point form of Ohm's Law  $\vec{J} = \sigma \vec{E}$
- Equation of continuity  $\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$
- $\vec{\nabla} \times \vec{J} = 0$
- Exponential decaying of charges  $\rho = \rho_0 e^{-\left(\frac{\sigma}{\epsilon}\right)t}$

## Boundary Condition

- $E_{1t} = E_{2t}$
- $\frac{J_{1t}}{\sigma_1} = \frac{J_{2t}}{\sigma_2}$
- $D_{1n} - D_{2n} = \epsilon_1 E_{1n} - \epsilon_2 E_{2n} = \epsilon_1 \frac{J_{1n}}{\sigma_1} - \epsilon_2 \frac{J_{2n}}{\sigma_2} = \rho_s$
- If  $\frac{\sigma_2}{\sigma_1} = \frac{\epsilon_2}{\epsilon_1}, \rho_s = 0$

## Resistance

- Resistivity  $\frac{1}{\sigma} [\Omega\text{-m}]$
- Resistance  $R = \frac{l}{\sigma S} = \text{resistivity} \times \frac{l}{S} [\Omega]$
- Leakage Resistance and Capacitor  $RC = \frac{\epsilon}{\sigma}$
- Power density  $\vec{E} \cdot \vec{J} = \frac{|\vec{J}|^2}{\sigma} [\frac{W}{m^3}]$

## Laplace's and Poisson's Equation

- Scalar Poisson's Equation:  $\nabla^2 V = -\frac{\rho}{\epsilon}$   
 Cartesian co-ordinate system,  $\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = -\frac{\rho}{\epsilon}$   
 Cylindrical co-ordinate system,  $\frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} = -\frac{\rho}{\epsilon}$
- Laplace's equation:  $\nabla^2 V = 0$  [If  $\rho$  is zero, i.e. a charge free area]
- Vector Poisson's Equation:  $\nabla^2 \vec{A} = -\mu \vec{J}$

### Case 1: Direct Integration

$$\vec{E}_r = -\vec{a}_r \frac{V_o}{r \ln \frac{b}{a}} \quad (\text{When } V_a = 0, V_b = V_o)$$

$$\vec{E}_r = \vec{a}_r \frac{V_o}{r \ln \frac{a}{b}} \quad (\text{When } V_a = V_o, V_b = 0)$$

### Case 2: Method of Image

$$V_p = \frac{Q}{4\pi\epsilon} \left( \frac{1}{\sqrt{x^2 + (y-d)^2 + z^2}} - \frac{1}{\sqrt{x^2 + (y+d)^2 + z^2}} \right) \quad (\text{Flat Reflection})$$

$$V_M = \frac{\rho l}{2\pi\epsilon} \ln \frac{a}{d} \quad (\text{Curved Reflection})$$

$$\vec{E}(x, y, z) = -\nabla V$$

$$C \approx \frac{\pi\epsilon}{\ln\left(\frac{D}{2a}\right)} [\text{F/m}], \text{ For } D \gg a \quad (\text{Gauss's Law})$$

$$C \approx \frac{\pi\epsilon}{\ln\left(\frac{D}{2a}\right) + \sqrt{\left(\frac{D}{2a}\right)^2 - 1}} [\text{F/m}], \text{ For } D \text{ is not much larger than conductor radius } a \quad (D=2h, \frac{1}{C} = \frac{1}{C_o} + \frac{1}{C_o})$$

### Case 3: Method of Separation of Variables (Electrostatic Boundary Value Problems)

| $k_x^2$  | $k_x$     | $X(x)$                                                                 |
|----------|-----------|------------------------------------------------------------------------|
| 0        | 0         | $A_0 x + B_0$                                                          |
| Positive | Real      | $C_1 e^{jk_x x} + C_2 e^{-jk_x x} = A_1 \sinh k_x x + B_1 \cosh k_x x$ |
| Negative | Imaginary | $D_1 e^{k_x x} + D_2 e^{-k_x x} = A_2 \sinh k_x x + B_2 \cosh k_x x$   |

$$V(x, y, z) = X(x)Y(y)Z(z)$$

$$\frac{1}{X(x)} \frac{d^2 X(x)}{dx^2} = -k_x^2$$

### Case 1: Sphere

$$B_{\text{inside}} = \frac{\mu_o r}{2\pi b^2} I$$

$$B_{\text{outside}} = \frac{\mu_o}{2\pi r} I$$

### Magnetic

$$\text{Magnetic flux (Total flux passing through a surface area)} \Phi = \int_s \vec{B} \cdot d\vec{s} = \oint_c \vec{A} \cdot d\vec{l}$$

$$\Phi = \oint_s \vec{B} \cdot d\vec{s} = 0$$

### Case 1: Transformer Action

$$EMF = -N \frac{d\Phi(t)}{dt}$$

### Case 2: Motional EMF

$$EMF = \oint_c (\vec{u} \times \vec{B}) \cdot d\vec{l}$$

### Case 3: Combined Transformer Action and Motional EMF

$$EMF = \oint_c (\vec{u} \times \vec{B}(t)) \cdot d\vec{l} - N \oint_s \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$$

### Ideal Transformer

$$\text{RMS value } V_s = 4.44 N_s f \Phi_m$$

$$\frac{V_p}{V_s} = \frac{N_p}{N_s} = \frac{I_s}{I_p}$$

**Case 1: Long solenoid**

- Find flux density from Ampere's Law

$$\oint_C \vec{B} d\vec{l} = \mu I_{enclosed} \rightarrow B l = \mu (n l) I$$

$$B = \mu n I$$

- Find flux and flux linkage

$$\Phi = \int_s \vec{B} \cdot d\vec{s} = B S = \mu n I \left( \frac{\pi d^2}{4} \right)$$

$$\lambda = n \Phi = \mu n^2 I \left( \frac{\pi d^2}{4} \right)$$

- Find Inductance

$$L = \frac{\lambda}{I} = \mu n^2 \left( \frac{\pi d^2}{4} \right)$$

**Magnetic Circuit Representation [When  $\mu$  is known and constant]**

$$\text{Reluctance } \mathfrak{R}_c = \frac{l_c}{\mu S_c}$$

**Graphic Solution for B and H [When  $\mu$  is not known and core is saturated]**

$$B_g = \frac{\mu_o}{l_g} N I - \frac{\mu_o l_c}{l_g} H_c$$

**Case 1: Toroidal Coil**

$$\mathfrak{R}_c = \frac{l_c}{\mu S_c} = \frac{2\pi r}{\mu(\pi r_o^2)}$$

$$\Phi = \frac{N I}{\mathfrak{R}}$$

**Case 2: Rectangular cross-sectional area toroid**

$$\Phi = \frac{\mu(N I) h}{2\pi} \ln \frac{b}{a}$$

$$L = \frac{\lambda}{I} = \frac{N \Phi}{I} = \frac{\mu N^2 h}{2\pi} \ln \frac{b}{a}$$

**Case 3: Inductance of a co-axial cable**

$$\text{Total Inductance} = L_{int} + L_{ext} = \frac{\mu_o}{8\pi} + \frac{\mu_o}{2\pi} \ln \frac{b}{a}$$

Skin effect: In high frequency applications, current is only at the surface,  $L_{int} = 0$

Thin conductor: Internal flux can be neglected

**Case 4: Inductance between 2 parallel cylindrical conductors carrying current in opposite directions**

$$\text{Total Inductance} = L_{int} + L_{ext} = \frac{\mu_o}{4\pi} + \frac{\mu_o}{\pi} \ln \frac{b}{a}$$

**Time-varying Magnetic Field (Current produces flux > Flux produces voltage)**

$$i(t) = I_m \cos \omega t$$

$$B(t) = B_m \cos \omega t$$

$$\Phi(t) = \varphi_m \cos \omega t$$

$$EMF = e(t) = \oint_C \vec{E}_B d\vec{l} = -N \frac{d\Phi}{dt} = -\frac{d\lambda}{dt}$$

$$\text{Relations of Electric and Magnetic field } \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

**Stored energy in a capacitor**

$$W_e = \frac{1}{2} C V^2 = \frac{1}{2} Q V [J]$$

$$w_e = \frac{1}{2} \epsilon E^2 = \frac{1}{2} \frac{D^2}{\epsilon} \left[ \frac{J}{m^3} \right]$$

**Stored energy in a magnetic field**

$$W_m = \frac{1}{2} L I^2 = \frac{1}{2} \lambda I [J]$$

$$w_m = \frac{1}{2\mu} B^2 = \frac{1}{2} \mu H^2 \left[ \frac{J}{m^3} \right]$$

**Core/ Iron Loss**

$$P_c = P_h + P_e = K_h f B_m^n + K_e f^2 B_m^2 \left[ \frac{W}{m^3} \right], K_e = \frac{\pi^2 \sigma}{6}$$

**Skin Effect**

$$\text{Skin Depth } \delta = \sqrt{\frac{2}{\sigma \mu \omega}} = \sqrt{\frac{1}{\sigma \mu \pi f}}$$

**Electrostatic Force**

$$\vec{F}_e = +\vec{\nabla} W_e |_{V \text{ invariant}} [N]$$

**Magnetic Force**

$$\vec{F}_m = \pm \vec{\nabla} W_m [N]$$

If a source is present (const I)  $\rightarrow \vec{F}_m$  is +ve gradient of  $W_m$

If no source to replenish the spend energy (const  $\lambda$ )  $\rightarrow \vec{F}_m$  is -ve gradient of  $W_m$

**Inductance**

$$L = \frac{N \Phi}{I} = \frac{\lambda}{I} = \frac{N^2}{\mathfrak{R}} [H/m]$$

$$L = \frac{d\lambda}{di} = \frac{N d\Phi}{di} [\text{If there is magnetic saturation}]$$

**Mutual flux and mutual inductance (M) [When the magnetic fields are coupling with each other]**

$$M_{12} = \frac{\lambda_{12}}{I_1} = \frac{N_2 \Phi_{12}}{I_1} [H]$$

$$\text{Self/ leakage inductance} = \frac{\lambda_1}{I_1} = \frac{N_1 \Phi_1}{I_1}$$

**Maxwell's Equation (Concept of displacement current)**

| Significance                                                                       | Original                                | Revised Differential                                                          | Revised Integral                                                                            |
|------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Faraday's Law                                                                      | $\vec{\nabla} \times \vec{E} = 0$       | $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$          | $\oint_C \vec{E} d\vec{l} = -N \frac{d\varphi}{dt}$                                         |
| Ampere's Law<br>( $\frac{\partial \vec{D}}{\partial t}$ is negligible in low freq) | $\vec{\nabla} \times \vec{H} = \vec{J}$ | $\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$ | $\oint_C \vec{H} d\vec{l} = I + \oint_S \frac{\partial \vec{D}}{\partial t} \cdot d\vec{s}$ |
| Gauss's Law                                                                        | $\vec{\nabla} \cdot \vec{D} = \rho$     | -                                                                             | -                                                                                           |
| No isolated magnetic pole                                                          | $\vec{\nabla} \cdot \vec{B} = 0$        | -                                                                             | -                                                                                           |

## Part B Formula Sheet

Transmission line equations:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$Z_L = Z_0 \frac{1 + \Gamma}{1 - \Gamma}$$

$$SWR = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l} \quad (\text{arbitrary load})$$

$$Z_{in} = jZ_0 \tan \beta l \quad (\text{short-circuited line})$$

$$Z_{in} = -jZ_0 \cot \beta l \quad (\text{open-circuited line})$$

Rectangular waveguide equations:

$$f_{cmn} = \frac{1}{2\pi\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

| Quantity    | $TE_{mn}$ Mode                                                                                      | $TM_{mn}$ Mode                                                                                           |
|-------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| $k$         | $\omega\sqrt{\mu\epsilon}$                                                                          | $\omega\sqrt{\mu\epsilon}$                                                                               |
| $k_c$       | $\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$                              | $\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$                                   |
| $\beta$     | $\sqrt{k^2 - k_c^2}$                                                                                | $\sqrt{k^2 - k_c^2}$                                                                                     |
| $\lambda_c$ | $\frac{2\pi}{k_c}$                                                                                  | $\frac{2\pi}{k_c}$                                                                                       |
| $E_z$       | 0                                                                                                   | $B_{mn} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}$                                       |
| $H_z$       | $A_{mn} \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}$                                  | 0                                                                                                        |
| $E_x$       | $\frac{j\omega\mu n\pi}{k_c^2 b} A_{mn} \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}$  | $\frac{-j\beta m\pi}{k_c^2 a} B_{mn} \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}$          |
| $E_y$       | $\frac{-j\omega\mu m\pi}{k_c^2 a} A_{mn} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}$ | $\frac{-j\beta n\pi}{k_c^2 b} B_{mn} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}$          |
| $H_x$       | $\frac{j\beta m\pi}{k_c^2 a} A_{mn} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}$      | $\frac{j\omega\epsilon n\pi}{k_c^2 b} B_{mn} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} e^{-j\beta z}$  |
| $H_y$       | $\frac{j\beta n\pi}{k_c^2 b} A_{mn} \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}$      | $\frac{-j\omega\epsilon m\pi}{k_c^2 a} B_{mn} \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{-j\beta z}$ |
| $Z$         | $Z_{TE} = \frac{k\eta}{\beta}$                                                                      | $Z_{TM} = \frac{\beta\eta}{k}$                                                                           |

**Two Smith charts will be provided by the examination section.**

**END OF EXAMINATION PAPER**